

Question 1: Define Evolution.

Any definition that clearly, unambiguously gives the sense of one of the following-

“..descent with modification...”

OR

“..the changes in the proportions of biological types in a population over generations..”

OR

“Biological evolution is change in the properties of groups of organisms over the course of generations...it embraces everything from slight changes in the proportions of different forms of a gene within a population to the alterations that led from the earliest organism to dinosaurs, bees, oaks, and humans”

Question 2: Differentiate between Micro- and Macro Evolution.

Any definition that gives the clear unambiguous sense of-

Micro: Evolution within populations

Macro: Evolution at higher scale typically resulting in the origin of new species or higher taxa.

Question 3: Differentiate between discrete and continuous traits.

	Discrete	Continuous
Loci:	One or few	Many
Environmental effect:	Very low	Large
Genotype-Phenotype map:	Simple, one-to-one	Complex, Many-to-many

Question 4: A single bacterium divides repeatedly to give rise to a colony of one million bacteria. Will all the bacteria be identical to each other? Explain.

No, they will not be perfectly identical.

When bacteria divide into daughter cells, bacteria must copy their DNA to pass it to daughter cells, a process called DNA replication.

This replication is not perfect, and random copy errors (mutations) occasionally occur in the genetic code.

Over many divisions, these accumulated errors ensure the colony consists of genetically diverse individuals.